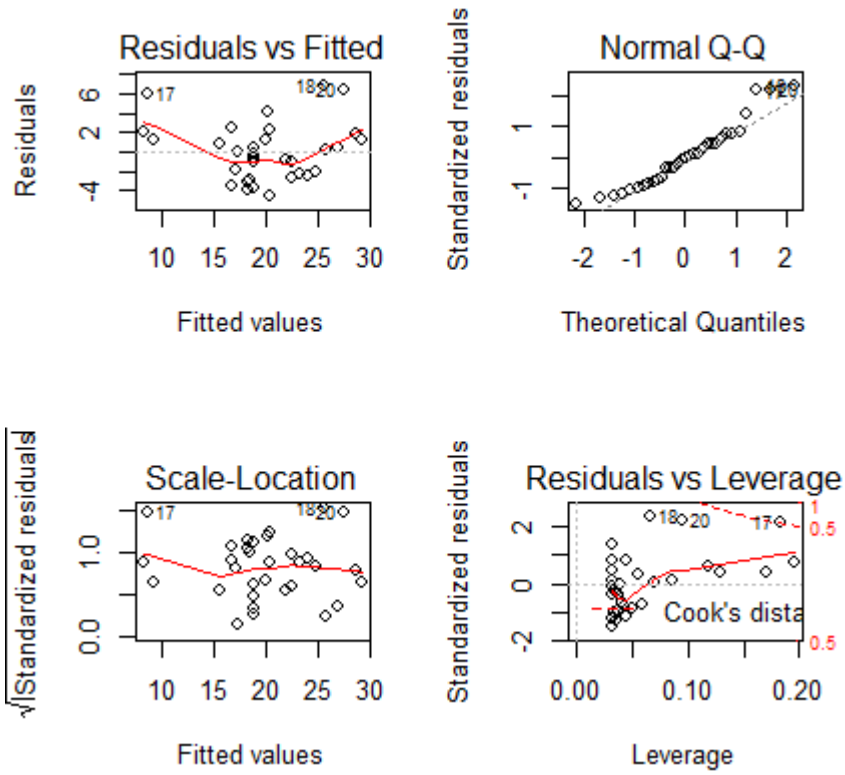
R Lab 5. REGRESSION DIAGNOSTICS

 # Run regression and look at residual plots. Split the plot window to get all the plots at once.

STUDENTIZED RESIDUALS AND OUTLIERS

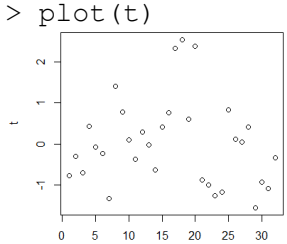
```
> attach(mtcars)
> reg = lm( mpg ~ wt )
> plot(wt,mpg)
> abline(reg)

> par(mfrow=c(2,2))
> plot(reg)
```

 # This command gives residual plots, and without "par", it gives them one at a time



Studentized residuals and testing for outliers

```
> t = rstudent(reg)          # Studentized residuals
> par(mfrow=c(1,1))          # Return to the 1x1 plot window
> plot(t)                    # See if there are any nonlinear trends
```



```
> t[ abs(t)>2 ]
      17       18       20
2.328162 2.537801 2.383844
```

 # Which of these residuals can be considered as outliers?
 # Compare with the Bonferroni-adjusted quantile from t-distribution.

```
> n = length(wt)
> qt( 0.025/n, n-2 )
[1] -3.478736

> t[ abs(t) > abs(qt( 0.025/n, n-2 )) ]
named numeric(0)
```

 # The test revealed no outliers.

 # Testing NORMALITY

```
> shapiro.test(t)

        Shapiro-Wilk normality test

data:  t
W = 0.92916, p-value = 0.03711
```

 # Rather marginal. Also look at the Normal Q-Q plot among residual plots above. It is not straight, so the data may be non-Normal. Shapiro-Wilk statistic W measures how close the graph is to a straight line.

 # Testing HOMOSCEDASTICITY (constant variance).
Look at the residual plots including a plot of t^2 vs fitted values.
Below is the Breausch-Pagan test, and it is included in R package called "car".

```
> install.packages("car")
> library("car")
> ncvTest(reg)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.03794177, Df = 1, p = 0.84556
```

 # This package also has a built-in outlier test

```
> outlierTest(reg)

No Studentized residuals with Bonferonni p < 0.05
Largest |rstudent|:
   rstudent unadjusted p-value Bonferonni p
18 2.537801           0.016788       0.5372
```

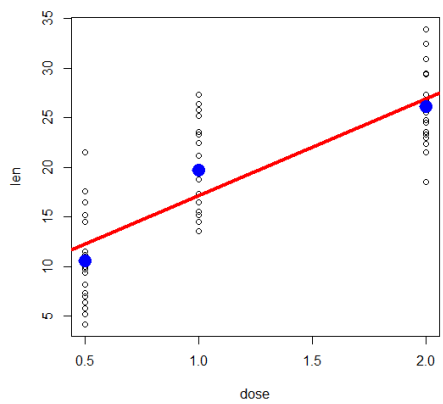
 # LACK OF FIT TEST

 # The ToothGrowth dataset has only 3 different values of X = dose

```
> attach(ToothGrowth)
> names(ToothGrowth)
[1] "len"  "supp" "dose"
> table(dose)
dose
0.5   1   2
 20  20  20
```

 # Fit two regression models:
 # reduced = simple linear regression predicting Y = length in terms of X = dose
 # full = using group means to predict Y for each value of X, thus treating X as a categorical variable

```
> reduced = lm(len ~ dose)
> full = lm(len ~ as.factor(dose))

> plot(dose,len)
> abline(reduced,col="red",lwd=4)
> points( dose, predict(full), col="blue", lwd=10 )
```



 # Here is the rigorous F-test for the lack of fit

```
> anova(reduced,full)
Analysis of Variance Table

Model 1: len ~ dose
Model 2: len ~ as.factor(dose)
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1     58 1227.9
2     57 1025.8  1    202.13 11.232 0.001432 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

 # Conclusion: the difference in SSreg is significant; the linear regression model does have a lack of fit.